AI ENGINEERING TOPIC CHECKLIST

All topics transcribed from the photos you sent • 5 areas × 20 topics = 100 topics

Only headings and topic names are included—no explanations or extra roadmap topics.

AREA 04 — EVALS

01 Golden Datasets	11 Regression Evals in CI
02 Error Analysis	12 A/B Testing Prompts
03 LLM as a Judge	13 Human Labeling
04 Pairwise Comparison	14 Synthetic Test Data
05 Rubric Scoring	15 Tracing
06 Faithfulness	16 Langfuse
07 Answer Relevance	17 OpenTelemetry for LLMs
08 Context Precision	18 P95 Latency
09 Tool Call Accuracy	19 Cost per Request
10 Trajectory Evals	20 User Feedback Loops

AREA 05 — AI SYSTEM DESIGN

01 Model Routing	11 PagedAttention
02 Model Fallbacks	12 Speculative Decoding
03 Semantic Caching	13 GPU vs API Cost
04 Prefix Caching	14 Batch Inference
05 Token Budgets	15 Async Job Pipelines
06 Streaming Responses	16 Provider Rate Limits
07 Time to First Token	17 Multi-tenant Design
08 Throughput vs Latency	18 RAG vs Fine-tuning
09 Inference Serving with vLLM	19 Context Window Management
10 Continuous Batching	20 Cost Optimization

AREA 06 — AI SECURITY

01 Prompt Injection	11 Secrets Management
02 Indirect Prompt Injection	12 API Key Rotation
03 Jailbreaks	13 Authorization for Agents
04 OWASP Top 10 for LLMs	14 Human Approval for Actions
05 Input Guardrails	15 Content Moderation
06 Output Validation	16 Data Poisoning
07 PII Redaction	17 Access Control in RAG
08 Data Leakage	18 MCP Server Trust
09 Least-privilege Tools	19 Audit Logging
10 Tool Sandboxing	20 Red Teaming

AREA 07 — DEPLOYMENT

01 Docker	11 Ollama for Local Models
02 Docker Compose	12 Autoscaling
03 Kubernetes Basics	13 Load Balancing
04 CI/CD Pipelines	14 Canary Releases
05 GitHub Actions	15 Feature Flags
06 Eval Gates in CI	16 Prompt Versioning
07 Environment Configs	17 Model Versioning
08 Infrastructure as Code	18 Health Checks
09 Serverless Functions	19 Alerts and Dashboards
10 GPU Instances	20 Rollbacks

AREA 08 — BACKEND

01 REST APIs	11 Caching with Redis
02 HTTP and Status Codes	12 Message Queues
03 Async and Await	13 Background Jobs
04 Streaming with SSE	14 Rate Limiting
05 WebSockets	15 Idempotency Keys
06 Authentication with JWT	16 Retries with Backoff
07 OAuth 2.0	17 File Uploads to S3
08 SQL vs NoSQL	18 Pagination
09 Database Indexing	19 Input Validation with Pydantic
10 Transactions and ACID	20 Logging and Monitoring